

APSPDCL · COMMERCIAL ROOFTOP

300 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

Net investment ₹1.16 Cr after input tax credit, generating 4,59,900 units in year one against an energy charge of ₹7.65 per unit.

PAYBACK 3.7 yrs Post-tax, undiscounted	PROJECT IRR 27.5% Unlevered, 30 years	COST OF SOLAR ENERGY ₹3.15 per kWh, against ₹7.65 from the grid	NET PRESENT VALUE ₹2.14 Cr Discounted at 10%
---	--	--	---

PLANT AND GENERATION	INVESTMENT AND TAX BASIS																																
<table><tr><td>Plant capacity</td><td>300 kWp</td></tr><tr><td>Year 1 generation</td><td>4,59,900 kWh</td></tr><tr><td>Specific yield</td><td>1,533 kWh / kWp / year</td></tr><tr><td>Supply category</td><td>HT Category I / II at 11 kV (HT supply at 11 kV)</td></tr><tr><td>Energy charge offset</td><td>₹7.65 / kWh</td></tr><tr><td>Units self-consumed</td><td>90% at the energy charge</td></tr><tr><td>Units exported</td><td>10% at ₹2.75 / kWh (APPC)</td></tr><tr><td>Year 1 gross saving</td><td>₹32.93 L</td></tr></table>	Plant capacity	300 kWp	Year 1 generation	4,59,900 kWh	Specific yield	1,533 kWh / kWp / year	Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)	Energy charge offset	₹7.65 / kWh	Units self-consumed	90% at the energy charge	Units exported	10% at ₹2.75 / kWh (APPC)	Year 1 gross saving	₹32.93 L	<table><tr><td>Price per watt-peak</td><td>₹40.50 / Wp (incl. GST)</td></tr><tr><td>Project price incl. GST</td><td>₹1.22 Cr</td></tr><tr><td>GST at 5% — recovered as ITC</td><td>₹5.79 L</td></tr><tr><td>Net capitalised cost</td><td>₹1.16 Cr</td></tr><tr><td>Depreciation block</td><td>40% WDV, Appendix I</td></tr><tr><td>Tax regime</td><td>s.115BAA — 25.168%</td></tr><tr><td>Annual maintenance</td><td>₹600 / kWp / year, year 1 free</td></tr><tr><td>30-year depreciation shield</td><td>₹29.12 L</td></tr></table>	Price per watt-peak	₹40.50 / Wp (incl. GST)	Project price incl. GST	₹1.22 Cr	GST at 5% — recovered as ITC	₹5.79 L	Net capitalised cost	₹1.16 Cr	Depreciation block	40% WDV, Appendix I	Tax regime	s.115BAA — 25.168%	Annual maintenance	₹600 / kWp / year, year 1 free	30-year depreciation shield	₹29.12 L
Plant capacity	300 kWp																																
Year 1 generation	4,59,900 kWh																																
Specific yield	1,533 kWh / kWp / year																																
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)																																
Energy charge offset	₹7.65 / kWh																																
Units self-consumed	90% at the energy charge																																
Units exported	10% at ₹2.75 / kWh (APPC)																																
Year 1 gross saving	₹32.93 L																																
Price per watt-peak	₹40.50 / Wp (incl. GST)																																
Project price incl. GST	₹1.22 Cr																																
GST at 5% — recovered as ITC	₹5.79 L																																
Net capitalised cost	₹1.16 Cr																																
Depreciation block	40% WDV, Appendix I																																
Tax regime	s.115BAA — 25.168%																																
Annual maintenance	₹600 / kWp / year, year 1 free																																
30-year depreciation shield	₹29.12 L																																

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,15,71,428	46,28,571	69,42,857	11,64,919
2	69,42,857	27,77,143	41,65,714	6,98,951
3	41,65,715	16,66,286	24,99,429	4,19,371
4	24,99,428	9,99,771	14,99,657	2,51,622
5	14,99,657	5,99,863	8,99,794	1,50,974
6	8,99,795	3,59,918	5,39,877	90,584
7	5,39,877	2,15,951	3,23,926	54,351
8	3,23,926	1,29,570	1,94,356	32,610
9	1,94,355	77,742	1,16,613	19,566
10	1,16,613	46,645	69,968	11,740
11	69,968	27,987	41,981	7,044
12	41,980	16,792	25,188	4,226
Tax shield, years 1-12				29,05,957

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

300 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	4,59,900	32,92,884	—	46,28,571	(3,36,166)	36,29,050	36,29,050
2	4,57,601	34,07,476	1,80,000	27,77,143	1,13,340	31,14,136	67,43,186
3	4,55,312	35,26,057	1,80,000	16,66,286	4,22,765	29,23,292	96,66,478
4	4,53,036	36,48,763	1,80,000	9,99,771	6,21,396	28,47,367	1,25,13,845
5	4,50,771	37,75,740	1,80,000	5,99,863	7,54,002	28,41,738	1,53,55,583
6	4,48,517	39,07,136	1,80,000	3,59,918	8,47,462	28,79,675	1,82,35,258
7	4,46,274	40,43,104	1,80,000	2,15,951	9,17,916	29,45,189	2,11,80,447
8	4,44,043	41,83,804	1,80,000	1,29,570	9,75,067	30,28,737	2,42,09,184
9	4,41,823	43,29,401	1,80,000	77,742	10,24,755	31,24,646	2,73,33,830
10	4,39,614	44,80,064	1,80,000	46,645	10,70,500	32,29,564	3,05,63,394
11	4,37,416	46,35,970	1,80,000	27,987	11,14,435	33,41,535	3,39,04,929
12	4,35,228	47,97,302	1,80,000	16,792	11,57,856	34,59,446	3,73,64,375
13	4,33,052	49,64,248	1,80,000	10,075	12,01,564	35,82,684	4,09,47,059
14	4,30,887	51,37,004	1,80,000	6,045	12,46,057	37,10,947	4,46,58,006
15	4,28,733	53,15,772	1,80,000	3,627	12,91,658	38,44,113	4,85,02,119
16	4,26,589	55,00,760	1,80,000	2,176	13,38,581	39,82,179	5,24,84,298
17	4,24,456	56,92,187	1,80,000	1,306	13,86,979	41,25,208	5,66,09,506
18	4,22,334	58,90,275	1,80,000	783	14,36,965	42,73,310	6,08,82,816
19	4,20,222	60,95,257	1,80,000	470	14,88,633	44,26,623	6,53,09,439
20	4,18,121	63,07,372	1,80,000	282	15,42,066	45,85,306	6,98,94,745
21	4,16,030	65,26,868	1,80,000	169	15,97,337	47,49,531	7,46,44,276
22	4,13,950	67,54,003	1,80,000	102	16,54,520	49,19,484	7,95,63,760
23	4,11,880	69,89,042	1,80,000	61	17,13,684	50,95,358	8,46,59,118
24	4,09,821	72,32,261	1,80,000	37	17,74,904	52,77,357	8,99,36,475
25	4,07,772	74,83,944	1,80,000	22	18,38,251	54,65,693	9,54,02,168
26	4,05,733	77,44,385	1,80,000	13	19,03,801	56,60,584	10,10,62,752
27	4,03,704	80,13,890	1,80,000	8	19,71,631	58,62,258	10,69,25,010
28	4,01,686	82,92,773	1,80,000	5	20,41,821	60,70,951	11,29,95,961
29	3,99,677	85,81,361	1,80,000	3	21,14,454	62,86,907	11,92,82,868
30	3,97,679	88,79,993	1,80,000	2	21,89,614	65,10,379	12,57,93,247
Total	1,28,41,861	16,94,29,096	52,20,000	1,15,71,425	3,84,15,848	12,57,93,247	12,57,93,247

RETURN METRICS

Simple payback	3.7 yrs
Discounted payback	4.9 yrs
Project IRR, unlevered	27.5%
Net present value at 10%	₹2.14 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.15 / kWh
Cumulative return, 30 years	987%
Total net cash flow, after tax	₹12.58 Cr
30-year generation	1,28,41,862 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.15
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.15 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 1,28,41,862 units, avoiding 10,530.3 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹1.22 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(iia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 300 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.